GITOPS / INFRASTRUCTURE AS CODE

GITOPS AND DECLARATIVE INFRASTRUCTURE

GitOps means managing infrastructure through Git as the single source of truth. The desired state of every system is declared in code, reviewed via pull request, and automatically reconciled by an agent. Configuration drift — the gap between what your system should be and what it is — becomes visible and automatically correctable.

KEY BENEFITS

- ✓ Configuration drift is detected and corrected automatically — no manual audits
- ✓ Every infrastructure change is reviewed, versioned, and instantly reversible
- ✓ Eliminates the 'works on prod but not in staging' class of incident
- ✓ Provides a complete, queryable history of every infrastructure change ever made

REAL-WORLD EXAMPLE

Weaveworks coined GitOps in 2017. Organisations adopting it consistently report fewer configuration-related incidents, faster MTTR, and dramatically improved audit trails.

PRACTICAL EXAMPLE

An engineer manually tweaks a Kubernetes resource limit directly on the cluster. ArgoCD detects the drift within 60 seconds, alerts the team, and reverts automatically. The correct fix goes through a reviewed PR.

REFLECTION QUESTION

Is your infrastructure's current state fully described in version-controlled code? Could you rebuild from scratch from that code alone?